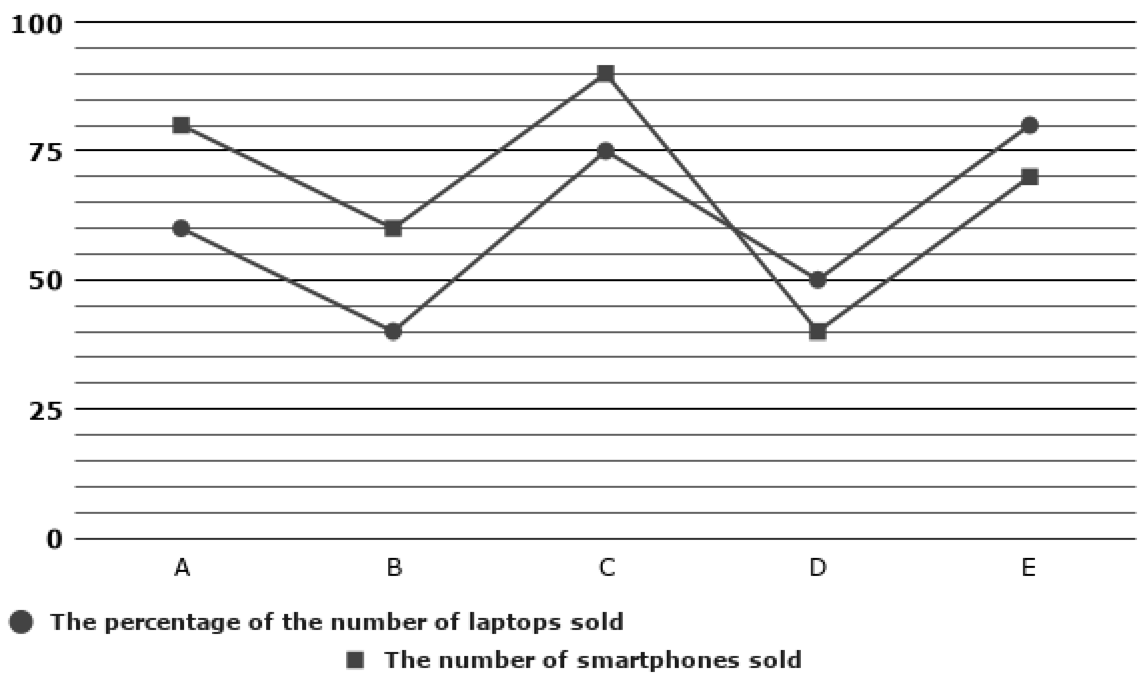
1. Questions

Study the following information carefully and answer the questions given below.

The given line graph shows the percentage of the number of laptops sold out of the total number of items sold, the number of smartphones sold by five different electronic companies (A, B, C, D, and E) in March 2026.

Note:

Total number of items sold by a company = The number of laptops sold + The number of smartphones sold



The number of smartphones sold by Company C is what percentage more or less than the number of smartphones sold by Company B?

a. 40% less

b. 50% more

c. 60% more

d. 45% less

e. 50% less

2. Questions

If the number of smartphones sold by Companies C and E increased by 20% and 30%, respectively in April from the previous month, find the total number of smartphones sold by these two companies together in April.

a. 185

b. 201
c. 199
d. 195
e. 211

3. Questions

What is the ratio of the number of smartphones sold by Company A to the number of smartphones sold by Company E?

a. 8 : 7
b. 9 : 7
c. 3 : 8
d. 4 : 7
e. 5 : 4

4. Questions

If the ratio of the number of laptops sold to the number of unsold laptops for Company A is 3:2, and the number of laptops sold by Company E is 70% of the total number of laptops manufactured by Company E, then find the total number of laptops unsold by Companies A and E together.

Note: The total number of laptops manufactured = (Number of laptops sold + Number of laptops unsold)

a. 180
b. 220
c. 200
d. 240
e. 160

5. Questions

Find the difference between the number of laptops sold by Companies B and C together and the total number of items sold by Company D.

a. 130
b. 310
c. 230
d. 170
e. 270

6. Questions

Study the following information carefully and answer the questions given below.

The table given below shows the total number of tourists visiting the five different destinations—P, Q, R, S, and T— recorded on a specific Monday in October 2026. The table also shows the respective ratio of the number of male tourists to female tourists for each destination.

Note: The total number of tourists who visited Destination = The total number of male tourists who visited Destination + The total number of female tourists who visited Destination

Destination	The total number of tourists	The ratio of the number of male to female tourists
P	1200	5 : 3
Q	1500	8 : 7
R	1800	5 : 4
S	960	7 : 5
T	1440	13 : 11

What is the total number of male tourists who visited Destinations P and R together on Monday?

a. 1650

b. 1800

c. 1950

d. 1500

e. 1750

7. Questions

The number of male tourists who visited Destination Q is what percentage of the total number of tourists who visited Destinations S and T together on Monday?

a. 25.00%

b. 33.33%

c. 40.00%

d. 50.00%

e. 20.00%

8. Questions

If the number of tourists who visited Destinations P and Q declined by 15% and 20%, respectively, from Monday, find the total number of tourists visiting these two destinations on Tuesday.

a. 2220

b. 1220

c. 2080

d. 1160

e. 1200

9. Questions

What is the respective ratio of the number of female tourists who visited Destinations S and T together to the total number of tourists who visited these two destinations together?

a. 47 : 120

b. 43 : 120

c. 61 : 130

d. 53 : 120

e. None of these

10. Questions

The number of female tourists who visited Destination Q is what percentage more or less than that of female tourists who visited Destination R?

a. 11.11% more

b. 12.5% less

c. 14.28% less

d. 12.5% more

e. 10% less

11. Questions

Study the following information carefully and answer the questions given below.

A magician holds two bags, Bag A and Bag B. The ratio of the number of Pink, Blue, and Purple balls in Bag A is 3:2:4, respectively, and Bag A contains a total of 45 balls. The number of Pink balls in Bag B is 20% more than the number of Pink balls in Bag A, and the ratio of the number of Blue balls to Purple balls in Bag B is 5:3. The total number of balls in Bag B is 50.

The number of Purple balls in Bag B is what percentage of the total number of Blue balls in both the bags together?

a. 30%

b. 45%

c. 40%

d. 50%

e. 25%

12. Questions

What is the difference between the total number of Pink balls in both bags together and the number of Purple balls in Bag A?

a. 11

b. 13

c. 15

d. 17

e. 9

13. Questions

If 5 more Blue balls are added to Bag A and 3 Pink balls are removed from Bag B, then find the new ratio of the number of Blue balls in Bag A to the number of Pink balls in Bag B.

a. 1 : 1

b. 2 : 3

c. 3 : 1

d. 4 : 5

e. 5 : 1

14. Questions

Ram and Raj start from Chennai and Bangalore towards each other. After crossing each other, Ram completes the remaining journey in 4.5 hours at 50 km/hr. If Raj's speed is 75 km/hr, find the time taken by Raj to cover the remaining journey.

a. 3 hours

b. 4 hours

c. 2.5 hours

d. 2 hours

e. 1.5 hours

15. Questions

The average age of four persons A, B, C, and D is 25 years. Person A is 4 years older than B who is 2 years younger than D. If A is 28 years old, find the age of C.

a. 26 years

b. 22 years

c. 28 years

d. 24 years

e. 21 years

16. Questions

20% of the bulbs are defective, and the remaining bulbs are sold at a 20% discount on the marked price. If the seller earns an overall profit of 28% on the total cost price of all the bulbs, then find the marked-up percentage.

a. 150%

b. 80%

c. 120%

d. 100%

e. 50%

17. Questions

A mixture contains 90 liters of alcohol and 30 liters of water. What quantity of the mixture should be replaced with water to obtain the ratio of alcohol and water equal to 1:1?

a. 30 liters

b. 45 liters

c. 50 liters

d. 40 liters

e. 25 liters

18. Questions

A and B start a business investing Rs. 4000 and Rs. 3000 respectively. A invests for 4 months and B invests for 8 months. The profit share of A is what percentage of the total profit of the business?

a. 40%

b. 50%

c. 60%

d. 45%

e. 35%

19. Questions

Arun is 4 times as efficient as Selva. Arun takes 30 days less than Selva to complete a work. Find the time in which they finish the work together.

a. 12 days
b. 10 days
c. 8 days
d. 15 days
e. 6 days

20. Questions

A sum of money placed at compound interest becomes four times of itself in 5 years. In how many years will it amount to 64 times of itself?

a. 15 years
b. 10 years
c. 20 years
d. 25 years
e. 30 years

21. Questions

What value will come in place of the question mark (?) in the following question?

$$\frac{?}{4} * \sqrt{2304} = \frac{192}{?} * 6^2$$

a. 24
b. 28
c. 16
d. 32
e. 20

22. Questions

$(0.09)^{18} * (0.027)^{14} \div (0.0081)^{12} = (0.3)^{?}$

a. 28
b. 30
c. 32
d. 26
e. 34

23. Questions

$(25^2 + 435 + 380) \div 40 = ? + 15$

a. 18
b. 25
c. 21
d. 24
e. 15

24. Questions

$(\sqrt{3136} + \sqrt{1024}) - (13 * 3) = ?^2$

a. 7
b. 9
c. 6
d. 8
e. 5

25. Questions

$(18^2 - \sqrt{1296}) \div (7^2 - 5^2) = ?$

a. 15
b. 14
c. 8
d. 12
e. 10

26. Questions

What approximate value will come in place of the question mark (?) in the following question?

$$\frac{83.99}{587.98} * \frac{13.99}{41.97} * ? = 16.01$$

a. 316
b. 336
c. 346
d. 326

e. 356

27. Questions

1727.98 ÷ 71.95 * 15.05 = ? + 24.99

a. 345
b. 335
c. 315
d. 325
e. 355

28. Questions

$$\frac{419.98}{6.99} - 13.98 * 15.02 + \sqrt{225.04} = ?$$

a. -125
b. -145
c. -135
d. -115
e. -155

29. Questions

119.97 * 8.03 = ? - 149.98 * 5.01

a. 1610
b. 1810
c. 1710
d. 1510
e. 1910

30. Questions

? + 7.02^3 = 1680.98 - 41.97

a. 986
b. 1296
c. 1216
d. 1196

e. 1396

31. Questions

What value will come in the place of (?) in the following number series?

232, ?, 243, 287, 254, 298

a. 276

b. 282

c. 264

d. 272

e. 268

32. Questions

13, 17, 26, 51, 100, ?

a. 235

b. 221

c. 215

d. 242

e. 210

33. Questions

22, 11, 11, ?, 33, 82.5

a. 16.5

b. 18.5

c. 15

d. 22

e. 12.5

34. Questions

635, 627, 600, 536, ?, 195

a. 420

b. 415

c. 405

d. 411

e. 418

35. Questions

14, 26, ?, 296, 1475

a. 68
b. 75
c. 82
d. 90
e. 72

36. Questions

Find the wrong term in the given series.

12, 16, 21, 33, 57, 105

a. 12
b. 16
c. 21
d. 33
e. 57

37. Questions

320, 160, 84, 40, 20, 10

a. 320
b. 160
c. 84
d. 40
e. 20

38. Questions

50, 51, 55, 64, 85, 105

a. 51
b. 55
c. 64
d. 85

e. 105

39. Questions

100, 101, 109, 136, 200, 330

a. 101
b. 109
c. 136
d. 200
e. 330

40. Questions

102, 98, 95, 90, 83, 72

a. 102
b. 98
c. 95
d. 90
e. 83

41. Questions

In each of the following questions, two equations are given. You have to solve both equations to find the relation between x and y.

I). $x^2 - 13x + 36 = 0$

II). $y^2 - 3y + 2 = 0$

a. x > y
b. x ≥ y
c. x < y
d. x ≤ y
e. x = y or the relationship can't be determined

42. Questions

I). $x^2 - 2x - 399 = 0$

II). $y^2 - 15y - 184 = 0$

a. x > y

b. $x \geq y$

c. $x < y$

d. $x \leq y$

e. $x = y$ or the relationship can't be determined

43. Questions

I). $x^2 + 33x + 242 = 0$

II). $y^2 + 52y + 660 = 0$

a. $x > y$

b. $x \geq y$

c. $x < y$

d. $x \leq y$

e. $x = y$ or the relationship can't be determined

44. Questions

I). $x^2 + 21x + 108 = 0$

II). $y^2 - 5y = 6^2$

a. $x > y$

b. $x \geq y$

c. $x < y$

d. $x \leq y$

e. $x = y$ or the relationship can't be determined

45. Questions

I). $3x + 12 = 27$

II). $y^2 - 13y + 40 = 0$

a. $x > y$

b. $x \geq y$

c. $x < y$

d. $x \leq y$

e. $x = y$ or the relationship can't be determined

Explanations:

1. Questions

Company A:

The percentage of laptops sold by Company A = 60%

The percentage of smartphones sold by Company A = 100% - 60% = 40%

The number of smartphones sold by Company A = 80

Therefore, 40% of the total items sold = 80

The total number of items sold by Company A = (80 / 40) x 100 = 200

The number of laptops sold by Company A = 200 - 80 = 120

Company B:

The percentage of laptops sold by Company B = 40%

The percentage of smartphones sold by Company B = 100% - 40% = 60%

The number of smartphones sold by Company B = 60

Therefore, 60% of the total items sold = 60

The total number of items sold by Company B = (60 / 60) x 100 = 100

The number of laptops sold by Company B = 100 - 60 = 40

Company C:

The percentage of laptops sold by Company C = 75%

The percentage of smartphones sold by Company C = 100% - 75% = 25%

The number of smartphones sold by Company C = 90

Therefore, 25% of the total items sold = 90

The total number of items sold by Company C = (90 / 25) x 100 = 360

The number of laptops sold by Company C = 360 - 90 = 270

Company D:

The percentage of laptops sold by Company D = 50%

The percentage of smartphones sold by Company D = 100% - 50% = 50%

The number of smartphones sold by Company D = 40

Therefore, 50% of the total items sold = 40

The total number of items sold by Company D = (40 / 50) x 100 = 80

The number of laptops sold by Company D = 80 - 40 = 40

Company E:

The percentage of laptops sold by Company E = 80%

The percentage of smartphones sold by Company E = 100% - 80% = 20%

The number of smartphones sold by Company E = 70

Therefore, 20% of the total items sold = 70

The total number of items sold by Company E = (70 / 20) x 100 = 350

The number of laptops sold by Company E = 350 - 70 = 280

Company	Total number of items Sold	Number of laptops Sold	Number of smartphones Sold
A	200	120	80
B	100	40	60
C	360	270	90
D	80	40	40
E	350	280	70

Answer: B

The number of smartphones sold by company C = 90

The number of smartphones sold by company B = 60

Required Percentage = [(90 - 60) / 60] * 100 = 30/60 * 100 = 50% more

Hence, the correct answer is option B.

2. Questions

Company A:

The percentage of laptops sold by Company A = 60%

The percentage of smartphones sold by Company A = 100% - 60% = 40%

The number of smartphones sold by Company A = 80

Therefore, 40% of the total items sold = 80

The total number of items sold by Company A = (80 / 40) x 100 = 200

The number of laptops sold by Company A = 200 - 80 = 120

Company B:

The percentage of laptops sold by Company B = 40%

The percentage of smartphones sold by Company B = 100% - 40% = 60%

The number of smartphones sold by Company B = 60

Therefore, 60% of the total items sold = 60

The total number of items sold by Company B = (60 / 60) x 100 = 100

The number of laptops sold by Company B = 100 - 60 = 40

Company C:

The percentage of laptops sold by Company C = 75%

The percentage of smartphones sold by Company C = 100% - 75% = 25%

The number of smartphones sold by Company C = 90

Therefore, 25% of the total items sold = 90

The total number of items sold by Company C = (90 / 25) x 100 = 360

The number of laptops sold by Company C = 360 - 90 = 270

Company D:

The percentage of laptops sold by Company D = 50%

The percentage of smartphones sold by Company D = 100% - 50% = 50%

The number of smartphones sold by Company D = 40

Therefore, 50% of the total items sold = 40

The total number of items sold by Company D = (40 / 50) x 100 = 80

The number of laptops sold by Company D = 80 - 40 = 40

Company E:

The percentage of laptops sold by Company E = 80%

The percentage of smartphones sold by Company E = 100% - 80% = 20%

The number of smartphones sold by Company E = 70

Therefore, 20% of the total items sold = 70

The total number of items sold by Company E = (70 / 20) x 100 = 350

The number of laptops sold by Company E = 350 - 70 = 280

Company	Total number of items Sold	Number of laptops Sold	Number of smartphones Sold
A	200	120	80
B	100	40	60
C	360	270	90
D	80	40	40
E	350	280	70

Answer: C

The number of smartphones sold by Company C in April = 120% of 90 = (120 / 100) * 90 = 108

The number of smartphones sold by Company E in April = 130% of 70 = (130 / 100) * 70 = 91

The total number of smartphones sold by Company C and Company E together in April = 108 + 91 = 199

Hence, the correct answer is option C.

3. Questions

Company A:

The percentage of laptops sold by Company A = 60%

The percentage of smartphones sold by Company A = 100% - 60% = 40%

The number of smartphones sold by Company A = 80

Therefore, 40% of the total items sold = 80

The total number of items sold by Company A = (80 / 40) x 100 = 200

The number of laptops sold by Company A = 200 - 80 = 120

Company B:

The percentage of laptops sold by Company B = 40%

The percentage of smartphones sold by Company B = 100% - 40% = 60%

The number of smartphones sold by Company B = 60

Therefore, 60% of the total items sold = 60

The total number of items sold by Company B = (60 / 60) x 100 = 100

The number of laptops sold by Company B = 100 - 60 = 40

Company C:

The percentage of laptops sold by Company C = 75%

The percentage of smartphones sold by Company C = 100% - 75% = 25%

The number of smartphones sold by Company C = 90

Therefore, 25% of the total items sold = 90

The total number of items sold by Company C = (90 / 25) x 100 = 360

The number of laptops sold by Company C = 360 - 90 = 270

Company D:

The percentage of laptops sold by Company D = 50%

The percentage of smartphones sold by Company D = 100% - 50% = 50%

The number of smartphones sold by Company D = 40

Therefore, 50% of the total items sold = 40

The total number of items sold by Company D = (40 / 50) x 100 = 80

The number of laptops sold by Company D = 80 - 40 = 40

Company E:

The percentage of laptops sold by Company E = 80%

The percentage of smartphones sold by Company E = 100% - 80% = 20%

The number of smartphones sold by Company E = 70

Therefore, 20% of the total items sold = 70

The total number of items sold by Company E = (70 / 20) x 100 = 350

The number of laptops sold by Company E = 350 - 70 = 280

Company	Total number of items Sold	Number of laptops Sold	Number of smartphones Sold
A	200	120	80
B	100	40	60
C	360	270	90
D	80	40	40
E	350	280	70

Answer: A

The number of smartphones sold by Company A = 80

The number of smartphones sold by Company E = 70

Required Ratio = 80 : 70

Dividing by 10:

Required Ratio = 8 : 7

Hence, the correct answer is option A.

4. Questions

Company A:

The percentage of laptops sold by Company A = 60%

The percentage of smartphones sold by Company A = 100% - 60% = 40%

The number of smartphones sold by Company A = 80

Therefore, 40% of the total items sold = 80

The total number of items sold by Company A = (80 / 40) x 100 = 200

The number of laptops sold by Company A = 200 - 80 = 120

Company B:

The percentage of laptops sold by Company B = 40%

The percentage of smartphones sold by Company B = 100% - 40% = 60%

The number of smartphones sold by Company B = 60

Therefore, 60% of the total items sold = 60

The total number of items sold by Company B = (60 / 60) x 100 = 100

The number of laptops sold by Company B = 100 - 60 = 40

Company C:

The percentage of laptops sold by Company C = 75%

The percentage of smartphones sold by Company C = 100% - 75% = 25%

The number of smartphones sold by Company C = 90

Therefore, 25% of the total items sold = 90

The total number of items sold by Company C = (90 / 25) x 100 = 360

The number of laptops sold by Company C = 360 - 90 = 270

Company D:

The percentage of laptops sold by Company D = 50%

The percentage of smartphones sold by Company D = 100% - 50% = 50%

The number of smartphones sold by Company D = 40

Therefore, 50% of the total items sold = 40

The total number of items sold by Company D = (40 / 50) x 100 = 80

The number of laptops sold by Company D = 80 - 40 = 40

Company E:

The percentage of laptops sold by Company E = 80%

The percentage of smartphones sold by Company E = 100% - 80% = 20%

The number of smartphones sold by Company E = 70

Therefore, 20% of the total items sold = 70

The total number of items sold by Company E = (70 / 20) x 100 = 350

The number of laptops sold by Company E = 350 - 70 = 280

Company	Total number of items Sold	Number of laptops Sold	Number of smartphones Sold
A	200	120	80
B	100	40	60
C	360	270	90
D	80	40	40
E	350	280	70

Answer: C

For Company A:

The ratio of laptops sold to unsold laptops for Company A = 3 : 2

The number of unsold laptops for Company A = (120 / 3) * 2 = 80

For Company E:

The total number of laptops manufactured by Company E = (280 / 70) * 100 = 400

The number of unsold laptops for Company E = 400 - 280 = 120

Total number of unsold laptops for Company A and Company E together = 80 + 120 = 200.

Hence, the correct answer is option C.

5. Questions

Company A:

The percentage of laptops sold by Company A = 60%

The percentage of smartphones sold by Company A = 100% - 60% = 40%

The number of smartphones sold by Company A = 80

Therefore, 40% of the total items sold = 80

The total number of items sold by Company A = (80 / 40) x 100 = 200

The number of laptops sold by Company A = 200 - 80 = 120

Company B:

The percentage of laptops sold by Company B = 40%

The percentage of smartphones sold by Company B = 100% - 40% = 60%

The number of smartphones sold by Company B = 60

Therefore, 60% of the total items sold = 60

The total number of items sold by Company B = (60 / 60) x 100 = 100

The number of laptops sold by Company B = 100 - 60 = 40

Company C:

The percentage of laptops sold by Company C = 75%

The percentage of smartphones sold by Company C = 100% - 75% = 25%

The number of smartphones sold by Company C = 90

Therefore, 25% of the total items sold = 90

The total number of items sold by Company C = (90 / 25) x 100 = 360

The number of laptops sold by Company C = 360 - 90 = 270

Company D:

The percentage of laptops sold by Company D = 50%

The percentage of smartphones sold by Company D = 100% - 50% = 50%

The number of smartphones sold by Company D = 40

Therefore, 50% of the total items sold = 40

The total number of items sold by Company D = (40 / 50) x 100 = 80

The number of laptops sold by Company D = 80 - 40 = 40

Company E:

The percentage of laptops sold by Company E = 80%

The percentage of smartphones sold by Company E = 100% - 80% = 20%

The number of smartphones sold by Company E = 70

Therefore, 20% of the total items sold = 70

The total number of items sold by Company E = (70 / 20) x 100 = 350

The number of laptops sold by Company E = 350 - 70 = 280

Company	Total number of items Sold	Number of laptops Sold	Number of smartphones Sold
A	200	120	80
B	100	40	60
C	360	270	90
D	80	40	40
E	350	280	70

Answer: C

The number of laptops sold by Companies B and C together = 40 + 270 = 310

The total number of items sold by Company D = 80

Required difference = 310 - 80 = 230

Hence, the correct answer is option C.

6. Questions

Destination P:

The total number of tourists who visited Destination P = 1200

The ratio of male to female tourists who visited Destination P = 5 : 3

The total number of male tourists who visited Destination P = 1200 * (5 / 8) = 750

The total number of female tourists who visited Destination P = 1200 * (3 / 8) = 450

Destination Q:

The total number of tourists who visited Destination Q = 1500

The ratio of male to female tourists who visited Destination Q = 8 : 7

The total number of male tourists who visited Destination Q = 1500 * (8 / 15) = 800

The total number of female tourists who visited Destination Q = 1500 * (7 / 15) = 700

Destination R:

The total number of tourists who visited Destination R = 1800

The ratio of male to female tourists who visited Destination R = 5 : 4

The total number of male tourists who visited Destination R = 1800 * (5 / 9) = 1000

The total number of female tourists who visited Destination R = 1800 * (4 / 9) = 800

Destination S:

The total number of tourists who visited Destination S = 960

The ratio of male to female tourists who visited Destination S = 7 : 5

The total number of male tourists who visited Destination S = 960 * (7 / 12) = 560

The total number of female tourists who visited Destination S = 960 * (5 / 12) = 400

Destination T:

The total number of tourists who visited Destination T = 1440

The ratio of male to female tourists who visited Destination T = 13 : 11

The total number of male tourists who visited Destination T = 1440 * (13 / 24) = 780

The total number of female tourists who visited Destination T = 1440 * (11 / 24) = 660

Final Value Table:

Destination	Total number of Tourists	Total number of male Tourists	Total number of female Tourists
P	1200	750	450
Q	1500	800	700
R	1800	1000	800
S	960	560	400
T	1440	780	660

Answer: E

The total number of male tourists who visited Destinations P and R together = (750 + 1000) = 1750

Hence, the correct answer is option E.

7. Questions

Destination P:

The total number of tourists who visited Destination P = 1200

The ratio of male to female tourists who visited Destination P = 5 : 3

The total number of male tourists who visited Destination P = 1200 * (5 / 8) = 750

The total number of female tourists who visited Destination P = 1200 * (3 / 8) = 450

Destination Q:

The total number of tourists who visited Destination Q = 1500

The ratio of male to female tourists who visited Destination Q = 8 : 7

The total number of male tourists who visited Destination Q = 1500 * (8 / 15) = 800

The total number of female tourists who visited Destination Q = 1500 * (7 / 15) = 700

Destination R:

The total number of tourists who visited Destination R = 1800

The ratio of male to female tourists who visited Destination R = 5 : 4

The total number of male tourists who visited Destination R = 1800 * (5 / 9) = 1000

The total number of female tourists who visited Destination R = 1800 * (4 / 9) = 800

Destination S:

The total number of tourists who visited Destination S = 960

The ratio of male to female tourists who visited Destination S = 7 : 5

The total number of male tourists who visited Destination S = 960 * (7 / 12) = 560

The total number of female tourists who visited Destination S = 960 * (5 / 12) = 400

Destination T:

The total number of tourists who visited Destination T = 1440

The ratio of male to female tourists who visited Destination T = 13 : 11

The total number of male tourists who visited Destination T = 1440 * (13 / 24) = 780

The total number of female tourists who visited Destination T = 1440 * (11 / 24) = 660

Final Value Table:

Destination	Total number of Tourists	Total number of male Tourists	Total number of female Tourists
P	1200	750	450
Q	1500	800	700
R	1800	1000	800
S	960	560	400
T	1440	780	660

Answer: B

The total number of tourists who visited Destination S and Destination T together = (960 + 1440) = 2400

The required percentage = (800 / 2400) * 100

The required percentage = 33.33%

Hence, the correct answer is option B.

8. Questions

Destination P:

The total number of tourists who visited Destination P = 1200

The ratio of male to female tourists who visited Destination P = 5 : 3

The total number of male tourists who visited Destination P = 1200 * (5 / 8) = 750

The total number of female tourists who visited Destination P = 1200 * (3 / 8) = 450

Destination Q:

The total number of tourists who visited Destination Q = 1500

The ratio of male to female tourists who visited Destination Q = 8 : 7

The total number of male tourists who visited Destination Q = 1500 * (8 / 15) = 800

The total number of female tourists who visited Destination Q = 1500 * (7 / 15) = 700

Destination R:

The total number of tourists who visited Destination R = 1800

The ratio of male to female tourists who visited Destination R = 5 : 4

The total number of male tourists who visited Destination R = 1800 * (5 / 9) = 1000

The total number of female tourists who visited Destination R = 1800 * (4 / 9) = 800

Destination S:

The total number of tourists who visited Destination S = 960

The ratio of male to female tourists who visited Destination S = 7 : 5

The total number of male tourists who visited Destination S = 960 * (7 / 12) = 560

The total number of female tourists who visited Destination S = 960 * (5 / 12) = 400

Destination T:

The total number of tourists who visited Destination T = 1440

The ratio of male to female tourists who visited Destination T = 13 : 11

The total number of male tourists who visited Destination T = 1440 * (13 / 24) = 780

The total number of female tourists who visited Destination T = 1440 * (11 / 24) = 660

Final Value Table:

Destination	Total number of Tourists	Total number of male Tourists	Total number of female Tourists
P	1200	750	450
Q	1500	800	700
R	1800	1000	800
S	960	560	400
T	1440	780	660

Answer: A

The total number of tourists who visited Destination P on Tuesday = 1200 * (85 / 100) = 1020

The total number of tourists who visited Destination Q on Tuesday = 1500 * (80 / 100) = 1200

The required total = 1020 + 1200 = 2220

Hence, the correct answer is option A.

9. Questions

Destination P:

The total number of tourists who visited Destination P = 1200

The ratio of male to female tourists who visited Destination P = 5 : 3

The total number of male tourists who visited Destination P = 1200 * (5 / 8) = 750

The total number of female tourists who visited Destination P = 1200 * (3 / 8) = 450

Destination Q:

The total number of tourists who visited Destination Q = 1500

The ratio of male to female tourists who visited Destination Q = 8 : 7

The total number of male tourists who visited Destination Q = 1500 * (8 / 15) = 800

The total number of female tourists who visited Destination Q = 1500 * (7 / 15) = 700

Destination R:

The total number of tourists who visited Destination R = 1800

The ratio of male to female tourists who visited Destination R = 5 : 4

The total number of male tourists who visited Destination R = 1800 * (5 / 9) = 1000

The total number of female tourists who visited Destination R = 1800 * (4 / 9) = 800

Destination S:

The total number of tourists who visited Destination S = 960

The ratio of male to female tourists who visited Destination S = 7 : 5

The total number of male tourists who visited Destination S = 960 * (7 / 12) = 560

The total number of female tourists who visited Destination S = 960 * (5 / 12) = 400

Destination T:

The total number of tourists who visited Destination T = 1440

The ratio of male to female tourists who visited Destination T = 13 : 11

The total number of male tourists who visited Destination T = 1440 * (13 / 24) = 780

The total number of female tourists who visited Destination T = 1440 * (11 / 24) = 660

Final Value Table:

Destination	Total number of Tourists	Total number of male Tourists	Total number of female Tourists
P	1200	750	450
Q	1500	800	700
R	1800	1000	800
S	960	560	400
T	1440	780	660

Answer: D

The total number of female tourists who visited Destination S and Destination T together = 400 + 660 = 1060

The total number of tourists who visited Destination S and Destination T together = 960 + 1440 = 2400

The required ratio = 1060 : 2400 = 106 : 240 = 53 : 120

Hence, the correct answer is option D.

10. Questions

Destination P:

The total number of tourists who visited Destination P = 1200

The ratio of male to female tourists who visited Destination P = 5 : 3

The total number of male tourists who visited Destination P = 1200 * (5 / 8) = 750

The total number of female tourists who visited Destination P = 1200 * (3 / 8) = 450

Destination Q:

The total number of tourists who visited Destination Q = 1500

The ratio of male to female tourists who visited Destination Q = 8 : 7

The total number of male tourists who visited Destination Q = 1500 * (8 / 15) = 800

The total number of female tourists who visited Destination Q = 1500 * (7 / 15) = 700

Destination R:

The total number of tourists who visited Destination R = 1800

The ratio of male to female tourists who visited Destination R = 5 : 4

The total number of male tourists who visited Destination R = 1800 * (5 / 9) = 1000

The total number of female tourists who visited Destination R = 1800 * (4 / 9) = 800

Destination S:

The total number of tourists who visited Destination S = 960

The ratio of male to female tourists who visited Destination S = 7 : 5

The total number of male tourists who visited Destination S = 960 * (7 / 12) = 560

The total number of female tourists who visited Destination S = 960 * (5 / 12) = 400

Destination T:

The total number of tourists who visited Destination T = 1440

The ratio of male to female tourists who visited Destination T = 13 : 11

The total number of male tourists who visited Destination T = 1440 * (13 / 24) = 780

The total number of female tourists who visited Destination T = 1440 * (11 / 24) = 660

Final Value Table:

Destination	Total number of Tourists	Total number of male Tourists	Total number of female Tourists
P	1200	750	450
Q	1500	800	700
R	1800	1000	800
S	960	560	400
T	1440	780	660

Answer: B

The number of female tourists who visited Destination Q = 700

The number of female tourists who visited Destination R = 800

The difference in the number of female tourists = 800 - 700 = 100

The required percentage = (100 / 800) * 100

The required percentage = 12.5% less

Hence, the correct answer is option B.

11. Questions

Total number of balls in Bag A = 45

The ratio of Pink, Blue, and Purple balls in Bag A = 3 : 2 : 4

The number of Pink balls in Bag A = (3 / 9) * 45 = 15

The number of Blue balls in Bag A = (2 / 9) * 45 = 10

The number of Purple balls in Bag A = (4 / 9) * 45 = 20

Total number of balls in Bag B = 50

The number of Pink balls in Bag B = 120% of 15 = (120 / 100) * 15 = 18

The number of Blue and Purple balls in Bag B = 50 - 18 = 32

The ratio of Blue to Purple balls in Bag B = 5 : 3

The number of Blue balls in Bag B = (5 / 8) * 32 = 20

The number of Purple balls in Bag B = (3 / 8) * 32 = 12

Let table

Bag	The number of Pink balls	The number of Blue balls	The number of Purple balls	The total number of balls
A	15	10	20	45
B	18	20	12	50

Answer: C

Total number of Blue balls in both bags together = 10 + 20 = 30

Required Percentage = (12 / 30) * 100 = 40%

Hence, the correct answer is option C.

12. Questions

Total number of balls in Bag A = 45

The ratio of Pink, Blue, and Purple balls in Bag A = 3 : 2 : 4

The number of Pink balls in Bag A = (3 / 9) * 45 = 15

The number of Blue balls in Bag A = (2 / 9) * 45 = 10

The number of Purple balls in Bag A = (4 / 9) * 45 = 20

Total number of balls in Bag B = 50

The number of Pink balls in Bag B = 120% of 15 = (120 / 100) * 15 = 18

The number of Blue and Purple balls in Bag B = 50 - 18 = 32

The ratio of Blue to Purple balls in Bag B = 5 : 3

The number of Blue balls in Bag B = (5 / 8) * 32 = 20

The number of Purple balls in Bag B = (3 / 8) * 32 = 12

Let table

Bag	The number of Pink balls	The number of Blue balls	The number of Purple balls	The total number of balls
A	15	10	20	45
B	18	20	12	50

Answer: B

The number of Pink balls in Bag A = 15

The number of Pink balls in Bag B = 18

Total number of Pink balls in both bags together = 15 + 18 = 33

The number of Purple balls in Bag A = 20

Required Difference = 33 - 20 = 13

Hence, the correct answer is option B.

13. Questions

Total number of balls in Bag A = 45

The ratio of Pink, Blue, and Purple balls in Bag A = 3 : 2 : 4

The number of Pink balls in Bag A = (3 / 9) * 45 = 15

The number of Blue balls in Bag A = (2 / 9) * 45 = 10

The number of Purple balls in Bag A = (4 / 9) * 45 = 20

Total number of balls in Bag B = 50

The number of Pink balls in Bag B = 120% of 15 = (120 / 100) * 15 = 18

The number of Blue and Purple balls in Bag B = 50 - 18 = 32

The ratio of Blue to Purple balls in Bag B = 5 : 3

The number of Blue balls in Bag B = (5 / 8) * 32 = 20

The number of Purple balls in Bag B = (3 / 8) * 32 = 12

Let table

Bag	The number of Pink balls	The number of Blue balls	The number of Purple balls	The total number of balls
A	15	10	20	45
B	18	20	12	50

Answer: A

The initial number of Blue balls in Bag A = 10

The new number of Blue balls in Bag A = 10 + 5 = 15

The initial number of Pink balls in Bag B = 18

The new number of Pink balls in Bag B = 18 - 3 = 15

Required Ratio = 15 : 15

Dividing by 15:

Required Ratio = 1 : 1

Hence, the correct answer is option A.

14. Questions

Answer: D

Let the speed of Ram be S1 and the speed of Raj be S2.

Let the time taken by Ram after crossing be T1 and the time taken by Raj be T2.

$S1 / S2 = \sqrt{(T2 / T1)}$

$50 / 75 = \sqrt{(T2 / 4.5)}$

$2 / 3 = \sqrt{(T2 / 4.5)}$

Squaring both sides of the equation.

$4 / 9 = T2 / 4.5$

$T2 = (4 * 4.5) / 9$

$T2 = 18 / 9$

Time taken by Raj = 2 hours

Hence, the correct answer is option D.

15. Questions

Answer: B

Total age of A, B, C, and D = 25 * 4 = 100 years

Age of person A = 28 years

Age of person B = 28 - 4 = 24 years

Age of person D = 24 + 2 = 26 years

Total age = Age of A + Age of B + Age of C + Age of D

100 = 28 + 24 + Age of C + 26

100 = 78 + Age of C

Age of C = 22 years

Hence, the correct answer is option B.

16. Questions

Answer: D

Let the total number of bulbs be 100 and the cost price of each bulb be 10.

Total cost price of bulbs = 100 * 10 = 1000

Total selling price of bulbs = 1000 * (128 / 100) = 1280

Number of defective bulbs = 20% of 100 = 20 bulbs

Number of sold bulbs = 100 - 20 = 80 bulbs

Selling price of each sold bulb = 1280 / 80 = 16

Selling price of bulb = Marked price of bulb * (80 / 100)

16 = Marked price of bulb * 0.8

Marked price of bulb = 16 / 0.8 = 20

Percentage marked over cost price = ((20 - 10) / 10) * 100 = 100%

Hence, the correct answer is option D.

17. Questions

Answer: D

Initial quantity of alcohol = 90 liters

Initial quantity of water = 30 liters

Total initial capacity = 120 liters

Let the replacement quantity be x liters.

Alcohol remaining = 90 - x * (90/120) = 90 - (3x/4)

Water remaining = 30 - x * (30/120) + x = 30 - (3x/4) + (4x/4) = 30 + (3x/4)

[90 - (3x/4)]/[30 + (3x/4)] = 1/1

90 - (3x/4) = 30 + (3x/4)

60 = 6x/4

240 = 6x

x = 40 liters

Hence, the correct answer is option D.

18. Questions

Answer: A

Time period for A = 4 months

Time period for B = 8 months

Profit ratio of A and B = (4000 * 4) : (3000 * 8)

Profit ratio = 16000 : 24000 = 2 : 3

Total parts of profit = 2 + 3 = 5 parts

Profit percentage of A = (2 / 5) * 100 = 40%

Hence, the correct answer is option A.

19. Questions

Answer: C

Efficiency ratio of Arun and Selva = 4 : 1

Time taken ratio of Arun and Selva = 1 : 4

Difference in time units = 4 - 1 = 3 units

3 units = 30 days

1 unit = 10 days

Time taken by Arun = 10 days

Total work = Efficiency of Arun * Time taken by Arun = 4 * 10 = 40 units

Combined efficiency = 4 + 1 = 5 units per day

Time taken together = 40 / 5 = 8 days

Hence, the correct answer is option C.

20. Questions

Answer: A

Let the principal amount be P.

The amount becomes 4 * P in 5 years.

The target amount is 64 * P.

64 can be written as 4 * 4 * 4.

Total time taken = 3 * 5 = 15 years

Hence, the correct answer is option A.

21. Questions

Answer: A

$\Rightarrow (? / 4) * \sqrt{2304} = (192 / ?) * 6^2$

$\Rightarrow (? / 4) * 48 = (192 / ?) * 36$

$\Rightarrow ? * 12 = 6912 / ?$

$\Rightarrow ?^2 = 6912 / 12$

$\Rightarrow ?^2 = 576$

$\Rightarrow ? = 24$

Hence, the correct answer is option A.

22. Questions

Answer: B

$\Rightarrow (0.09)^{18} * (0.027)^{14} \div (0.0081)^{12} = (0.3)^?$

$\Rightarrow (0.3^2)^{18} * (0.3^3)^{14} \div (0.3^4)^{12} = (0.3)^?$

$\Rightarrow 0.3^{36} * 0.3^{42} \div 0.3^{48} = (0.3)^?$

$\Rightarrow 0.3^{(36 + 42 - 48)} = (0.3)^?$

$\Rightarrow 0.3^{30} = (0.3)^?$

$\Rightarrow ? = 30$

Hence, the correct answer is option B.

23. Questions

Answer: C

$\Rightarrow (25^2 + 435 + 380) \div 40 = ? + 15$

$\Rightarrow (625 + 435 + 380) \div 40 = ? + 15$

$\Rightarrow 1440 \div 40 = ? + 15$

$\Rightarrow 36 = ? + 15$

$\Rightarrow ? = 36 - 15$

$\Rightarrow ? = 21$

Hence, the correct answer is option C.

24. Questions

Answer: A

$\Rightarrow (\sqrt{3136} + \sqrt{1024}) - (13 * 3) = ?^2$

$\Rightarrow (56 + 32) - 39 = ?^2$

$\Rightarrow 88 - 39 = ?^2$

$\Rightarrow 49 = ?^2$

$\Rightarrow ? = 7$

Hence, the correct answer is option A.

25. Questions

Answer: D

$\Rightarrow (18^2 - \sqrt{1296}) \div (7^2 - 5^2) = ?$

$\Rightarrow (324 - 36) \div (49 - 25) = ?$

$\Rightarrow 288 \div 24 = ?$

$\Rightarrow ? = 12$

Hence, the correct answer is option D.

26. Questions

Answer: B

⇒ (83.99 / 587.98) * (13.99 / 41.97) * ? = 16.01

⇒ (84 / 588) * (14 / 42) * ? = 16

⇒ (1 / 7) * (1 / 3) * ? = 16

⇒ (1 / 21) * ? = 16

⇒ ? = 16 * 21

⇒ ? = 336

Hence, the correct answer is option B.

27. Questions

Answer: B

⇒ 1727.98 ÷ 71.95 * 15.05 = ? + 24.99

⇒ 1728 ÷ 72 * 15 = ? + 25

⇒ 24 * 15 = ? + 25

⇒ 360 = ? + 25

⇒ ? = 360 - 25

⇒ ? = 335

Hence, the correct answer is option B.

28. Questions

Answer: C

⇒ (419.98 / 6.99) - 13.98 * 15.02 + √225.04 = ?

⇒ (420 / 7) - 14 * 15 + √225 = ?

⇒ 60 - 14 * 15 + 15 = ?

⇒ 60 - 210 + 15 = ?

⇒ 75 - 210 = ?

⇒ ? = -135

Hence, the correct answer is option C.

29. Questions

Answer: C

⇒ 119.97 * 8.03 = ? - 149.98 * 5.01

⇒ 120 * 8 = ? - 150 * 5

⇒ 960 = ? - 750

⇒ ? = 960 + 750

⇒ ? = 1710

Hence, the correct answer is option C.

30. Questions

Answer: B

⇒ ? + 7.02^3 = 1680.98 - 41.97

⇒ ? + 7^3 = 1681 - 42

⇒ ? + 343 = 1681 - 42

⇒ ? + 343 = 1639

⇒ ? = 1639 - 343

⇒ ? = 1296

Hence, the correct answer is option B.

31. Questions

Answer: A

The series follows the pattern of alternate addition and subtraction:

232 + 44 = **276**

276 - 33 = 243

243 + 44 = 287

287 - 33 = 254

254 + 44 = 298

So, ? = 276.

Hence, the correct answer is option A.

32. Questions

Answer: B

The series follows the pattern of addition of squares of consecutive prime numbers:

13 + 2^2 = 17

17 + 3^2 = 26

26 + 5^2 = 51

51 + 7^2 = 100

$100 + 11^2 = $ **221**

So, ? = 221.

Hence, the correct answer is option B.

33. Questions

Answer: A

The series follows the pattern of multiplication by increasing multiples of 0.5:

22 * 0.5 = 11

11 * 1 = 11

11 * 1.5 = **16.5**

16.5 * 2 = 33

33 * 2.5 = 82.5

So, ? = 16.5.

Hence, the correct answer is option A.

34. Questions

Answer: D

The series follows the pattern of subtraction of cubes of consecutive numbers:

$635 - 2^3 = 627$

$627 - 3^3 = 600$

$600 - 4^3 = 536$

$536 - 5^3 = $ **411**

$411 - 6^3 = 195$

So, ? = 411.

Hence, the correct answer is option D.

35. Questions

Answer: B

The series follows the pattern of multiplying by a number and subtracting the same number:

14 * 2 - 2 = 26

26 * 3 - 3 = **75**

75 * 4 - 4 = 296

296 * 5 - 5 = 1475

So, ? = 75.

Hence, the correct answer is option B.

36. Questions

Answer: B

⇒ 12 + 3 = 15 (Therefore, 16 is wrong and should be 15)

⇒ 15 + 6 = 21

⇒ 21 + 12 = 33

⇒ 33 + 24 = 57

⇒ 57 + 48 = 105

⇒ Logic: Differences are in geometric progression (3, 6, 12, 24, 48).

Hence, the correct answer is option B.

37. Questions

Answer: C

⇒ 320 / 2 = 160

⇒ 160 / 2 = 80 (Therefore, 84 is wrong and should be 80)

⇒ 80 / 2 = 40

⇒ 40 / 2 = 20

⇒ 20 / 2 = 10

⇒ Logic: Each term is divided by 2 to get the next term.

Hence, the correct answer is option C.

38. Questions

Answer: D

⇒ 50 + 1 = 51

⇒ 51 + 4 = 55

⇒ 55 + 9 = 64

⇒ 64 + 16 = 80 (Therefore, 85 is wrong and should be 80)

⇒ 80 + 25 = 105

⇒ Logic: Differences are squares of natural numbers (1, 4, 9, 16, 25).

Hence, the correct answer is option D.

39. Questions

Answer: E

⇒ 100 + 1 = 101

⇒ 101 + 8 = 109

⇒ 109 + 27 = 136

⇒ 136 + 64 = 200

⇒ 200 + 125 = 325 (Therefore, 330 is wrong and should be 325)

⇒ Logic: Differences are cubes of natural numbers (1, 8, 27, 64, 125).

Hence, the correct answer is option E.

40. Questions

Answer: A

⇒ 100 - 2 = 98 (Therefore, 102 is wrong and should be 100)

⇒ 98 - 3 = 95

⇒ 95 - 5 = 90

⇒ 90 - 7 = 83

⇒ 83 - 11 = 72

⇒ Logic: Differences are consecutive prime numbers (2, 3, 5, 7, 11).

Hence, the correct answer is option A.

41. Questions

Answer: A

I). $x^2 - 13x + 36 = 0$

$x^2 - 4x - 9x + 36 = 0$

$x(x - 4) - 9(x - 4) = 0$

$(x - 4)(x - 9) = 0$

$x = 4, 9$

II). $y^2 - 3y + 2 = 0$

$y^2 - 1 - 2y + 2 = 0$

$y(y - 1) - 2(y - 1) = 0$

$(y - 1)(y - 2) = 0$

$y = 1, 2$

Comparison:

4 > 1

$4 > 2$

$9 > 1$

$9 > 2$

$x > y$

Hence, the correct answer is option A.

42. Questions

Answer: E

I). $x^2 - 2x - 399 = 0$

$x^2 + 19x - 21x - 399 = 0$

$x(x + 19) - 21(x + 19) = 0$

$(x + 19)(x - 21) = 0$

$x = -19, 21$

II). $y^2 - 15y - 184 = 0$

$y^2 + 8y - 23y - 184 = 0$

$y(y + 8) - 23(y + 8) = 0$

$(y + 8)(y - 23) = 0$

$y = 23, -8$

Comparison:

$-19 < 23$

$-19 < -8$

$21 < 23$

$21 > -8$

The relationship can’t be determined.

Hence, the correct answer is option E.

43. Questions

Answer: B

I). $x^2 + 33x + 242 = 0$

$x^2 + 11x + 22x + 242 = 0$

$x(x + 11) + 22(x + 11) = 0$

$(x + 11)(x + 22) = 0$

x = -11, -22

II). $y^2 + 52y + 660 = 0$

$y^2 + 22y + 30y + 660 = 0$

$y(y + 22) + 30(y + 22) = 0$

$(y + 22)(y + 30) = 0$

y = -22, -30

Comparison:

-11 > -22

-11 > -30

-22 = -22

-22 > -30

$x \geq y$

Hence, the correct answer is option B.

44. Questions

Answer: C

I). $x^2 + 21x + 108 = 0$

$x^2 + 9x + 12x + 108 = 0$

$x(x + 9) + 12(x + 9) = 0$

$(x + 9)(x + 12) = 0$

x = -9, -12

II). $y^2 - 5y = 6^2$

$y^2 - 5y - 36 = 0$

$y^2 - 9y + 4y - 36 = 0$

$y(y - 9) + 4(y - 9) = 0$

$(y - 9)(y + 4) = 0$

y = 9, -4

Comparison:

-9 < 9

-9 < -4

$-12 < 9$

$-12 < -4$

$x < y$

Hence, the correct answer is option C.

45. Questions

Answer: D

I). $3x + 12 = 27$

$3x = 15$

$x = 5$

II). $y^2 - 13y + 40 = 0$

$y^2 - 5y - 8y + 40 = 0$

$y(y - 5) - 8(y - 5) = 0$

$y = 5, 8$

Comparison:

$5 = 5$

$5 < 8$

$x \leq y$

Hence, the correct answer is option D.